Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/13/2026
PI Name:	Varanoske, Alyssa Ph.D.		
Project Title:	Exploration Exercise Treadmill Requirements		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **EVA**:Risk of Mission Impacting Injury and Compromised Performance and Long-Term Health Effects due to EVA Operations (3) **Muscle and Aerobic**:Risk of Mission Performance Decrements and Adverse Health Outcomes Resulting from Reduced Cardiorespiratory and Musculoskeletal Fitness (4) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	
Organization Name:	KBR/NASA Johnson Space Center		
PI Address 1:	NASA Johnson Space Center		
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PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	06/27/2022	**End Date:**	09/30/2034
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Project updated due to P.I. change to A. Varanoske. Previous PI: K. Marshall-Goebel. (Ed. 7/25/24) NOTE: Project start date and end date updated due to P.I. change to K. Marshall-Goebel and selection for flight. Original end date: 09/30/2027. Information per T. Finkle/NASA-JSC. (Ed. 8/4/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sibonga, Jean (NASA Johnson Space Center) Smith, Scott Ph.D. (NASA Johnson Space Center) Zwart, Sara Ph.D. (NASA Johnson Space Center) Spector, Elizabeth (KBR/NASA Johnson Space Center) Peters, Brian (KBR/NASA Johnson Space Center) Prejean, Brian (KBR/NASA Johnson Space Center) Young, Millenia (NASA Johnson Space Center) Marshall-Goebel, Karina (NASA Johnson Space Center)		

Grant/Contract No.:	Directed Research
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Exploration class missions including Artemis and beyond are required to have an exercise device that is lightweight and has a small footprint. This has resulted in the development of exercise devices that are motorized and flywheel-based to provide both aerobic and resistive training on one platform. These devices provide a variety of full body resistance exercise options as well as aerobic rowing and cycling, but no treadmill. Because a treadmill has been available for use on the International Space Station (ISS) since 2001, we do not understand what the effect of removing the treadmill will have on muscle performance, aerobic fitness, bone health, sensorimotor performance, or psychological health. Specific Aims: 1) We aim to quantify the effect of cycle ergometry and resistive exercise only, throughout an entire spaceflight mission on muscle performance, aerobic fitness, bone health, sensorimotor performance, or psychological health by removing the treadmill portion of exercise on ISS and replacing it with additional time on the ISS cycle ergometer (CEVIS), as prescribed by the astronaut trainers, while keeping exercise on the Advanced Resistive Exercise Device (ARED) the same. 2) We aim to quantify the effectiveness of the European Enhanced Exploration Exercise Device (E4D) during an entire spaceflight mission on bone, muscle, aerobic, and sensorimotor health and performance by having crewmembers utilize the E4D exclusively.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The major aim of this study is to provide exploration vehicle system mass and volume allocation requirements for a Mars exercise hardware capability and inform NASA/commercial developers for exercise requirements to effectively prevent physiological deconditioning during long-duration spaceflight. Specifically, if results show that a treadmill is a necessary component of the in-flight exercise training regime to prevent physiological deconditioning and ensure crew readiness, development of new exercise hardware that encompasses a treadmill may be necessary. This future work may require establishing partnerships with private industry, academia, or other governmental organizations to develop additional exercise capabilities and continue research to advance knowledge in this technical area.
Task Progress:	In this reporting period, data collection on Active (CEVIS+ARED) and Control (nominal International Space Station/ISS exercise: T2, ARED, and CEVIS) subjects has continued. To date, five Active subjects (CEVIS+ARED) have been enrolled in the study, and data collection has been completed on all five of these subjects (an additional two completers from the last report). Sixteen Control subjects have been enrolled (data share only), and data collection has been completed on ten of these subjects. Recruitment for all groups, including both Active groups (CEVIS+ARED group and E4D group) and the Control group, is underway. Although the E4D is not yet available for use on the ISS, it is anticipated to launch this spring, and data collection in the Active E4D group is expected to begin this FY. Informed consent briefings have been presented to the crew for two separate missions throughout this reporting period. Additionally, changes have been made to the study protocol documents and informed consent to reflect updates to Med Ops testing, including the removal of isokinetic testing pre- and post-flight and removal of in-flight metabolic measures for maximal aerobic capacity testing. These have since been added as additional research measures in Active subjects. Additionally, in-flight urine collection is no longer being collected in Active participants, and pre- and post-flight urine collection has been modified to include collection over a 24-hour span rather than a 48-hour span. In-flight isometric mid-thigh pull (IMTP) testing has been increased from 20 min to 40 min, and all sessions are now being remotely guided by laboratory support staff. Details for maximal allowances for device utilization have been included in the study consent and associated documents. The primary funding agency is transitioning from the Mars Campaign Office (MCO) to the Human Research Program (HRP). A new Experiment Document (ED) and Research Data Submission Agreement (RDSA) have been published.
Bibliography Type:	Description: (Last Updated: 04/15/2026)
Abstracts for Journals and Proceedings	Varanoske AN, Prejean BJ, Strock NC, Conly D, Peters BT, Gardner DN, Cooper LL, Sibonga JD, Smith SM, Zwart SR, Spector ER, Young M, Marshall-Goebel K. "Effects of replacing treadmill running with alternative exercise countermeasures during long-duration spaceflight on astronaut health and performance." 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026